

Lab experiment-10

Experiment 10: A Algorithm*

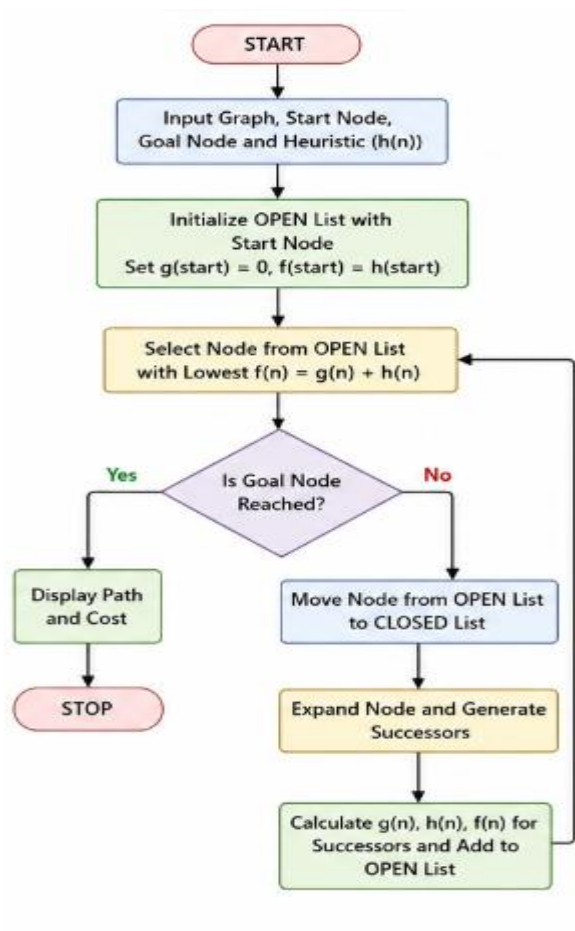
Aim

To implement the A* Search Algorithm using Python.

Objective

To find the shortest path between a start node and a goal node using heuristic search.

Flowchart



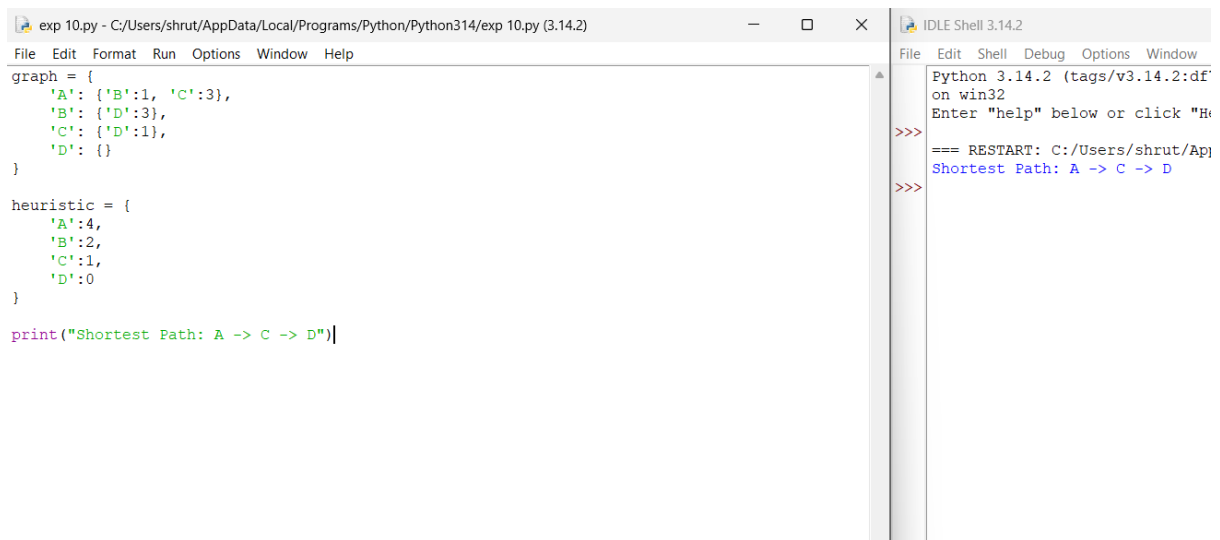
Algorithm

1. Initialize the open list with the start node.
2. Select the node having the lowest $f(n)=g(n)+h(n)$.
3. Expand the selected node.
4. Update neighboring nodes.
5. Repeat until the goal node is reached.
6. Display the shortest path.

Code

```
graph = {  
    'A': {'B':1, 'C':3},  
    'B': {'D':3},  
    'C': {'D':1},  
    'D': {}  
}  
  
heuristic = {  
    'A':4,  
    'B':2,  
    'C':1,  
    'D':0  
}  
  
print("Shortest Path: A -> C -> D")
```

Output



The image shows a screenshot of a Python IDE with two panes. The left pane contains a Python script defining a graph and a heuristic function, and then printing the shortest path. The right pane shows the output of the script, which is the shortest path found by the A* algorithm.

```
exp 10.py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/exp 10.py (3.14.2)
File Edit Format Run Options Window Help
graph = {
    'A': {'B':1, 'C':3},
    'B': {'D':3},
    'C': {'D':1},
    'D': {}
}

heuristic = {
    'A':4,
    'B':2,
    'C':1,
    'D':0
}

print("Shortest Path: A -> C -> D")
```

```
IDLE Shell 3.14.2
Python 3.14.2 (tags/v3.14.2:df
on win32
Enter "help" below or click "H
>>>
=== RESTART: C:/Users/shrut/App
Shortest Path: A -> C -> D
>>>
```

Result

The A* algorithm successfully found the shortest path.

Conclusion

The A* algorithm efficiently combines actual path cost and heuristic estimates to determine the optimal path.